

PRASAD MK

Solution Architect · Technical Lead · Full Stack SRE · Cloud Infrastructure & AI-Driven Operations

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SUMMARY

Technologist with vast experience in building and operating large-scale distributed systems, including 8+ years as the primary SRE owner for a multi-tenant SaaS platform sustaining 99.99% uptime across the Americas. Background spans C/C++ and Go systems programming, Java/Spring Boot services, and reliability engineering for infrastructure processing 8TB/day through Kafka and Pulsar pipelines. Led a Kubernetes migration that cut annual infrastructure cost by \$2.1M and collaborated across product, engineering, and support teams to drive SLO adoption at enterprise scale. Currently a Solutions Architect / SRE on a 70,000-user financial services platform at Wipro/PNC, with a track record of resolving 100% of Fortune 500 production escalations at Apple, Samsung, Disney, FedEx, and PG&E.

TECHNICAL SKILLS

Languages	Go, C/C++, Java (Spring Boot), Python, Bash, Perl
Distributed Systems	Large-scale distributed systems design, capacity planning, failover design, high-throughput data pipelines, multi-region SaaS architecture
Reliability / SRE	SLO/SLI ownership, on-call incident management, root-cause analysis, disaster recovery, runbook-as-code, 99.99% uptime ownership
Cloud & IaC	AWS (EKS, EC2, VPC, S3, RDS, Lambda, Bedrock), GCP, Azure, Terraform, Ansible, CloudFormation
Containers & CI/CD	Kubernetes, OpenShift, Helm, Docker, ArgoCD, Jenkins, GitLab CI, GitHub Actions
Observability	Dynatrace, Prometheus, Grafana, ELK Stack, PagerDuty
Data & Streaming	Kafka, Pulsar, Kafka Connect, PostgreSQL, MySQL, DynamoDB
AI & Agentic Systems	LLMs, AWS Bedrock, multi-agent orchestration, RAG pipelines, LangChain, Vector Databases, Voice AI integration

EXPERIENCE

Wipro Ltd. — Client: PNC Bank

Oct 2025 – Present

Solutions Architect Site Reliability Engineer ·

Pittsburgh, PA

- Architected a microservices-based Alert Management System on OpenShift/Kubernetes with a full observability stack, defining SLO targets and automated alerting that detects degradation in under 5 minutes across 70,000 enterprise users.
- Built backend services in Java/Spring Boot with 100% JUnit/Mockito coverage on critical paths; led CI/CD pipeline development via Jenkins and Terraform with integrated security scanning, achieving zero critical vulnerabilities in production.
- Built agentic AI backend services on AWS Bedrock with multi-agent orchestration and RAG pipelines for internal automation tooling; patent pending on the reputation-sorting agent design.
- Implemented runbook-as-code and platform-as-code provisioning standards, partnering across product, security, and application teams so on-call engineers can diagnose and remediate incidents without escalation.

Kakudi

Jun 2025 – Sep 2025

Software Engineer – Cloud & AI Systems ·

Folsom, CA

- Designed and deployed cloud-native microservices on Kubernetes using Terraform-provisioned AWS infrastructure; built Python automation and Ansible playbooks for operational workflows.
- Built agentic AI systems on AWS Bedrock with multi-agent orchestration, RAG pipelines, and CI/CD for automated model deployment; developed Voice AI agents integrating live data sources for nonprofit and real-estate clients.

OpenText

Feb 2023 – May 2025

Site Reliability Engineer ·

Rocklin, CA

- Sole SRE owner for SaaS AIOps and Network Operations Manager deployments across the Americas; maintained 99.99% SLA uptime, owned SOC2 compliance and disaster recovery strategy, and cut incident resolution time by 45% through Linux kernel diagnostics and packet-level tracing.
- Led migration of 50+ microservices to Kubernetes/OpenShift, compressing multi-tenant provisioning from days to 30 minutes and cutting annual cloud infrastructure cost by \$2.1M through right-sizing and Terraform automation.
- Built a Python-based ML cloud-cost optimizer using AWS Cost Explorer APIs, cutting quarterly cloud spend by 20%, and developed Go-based metric collectors pulling performance data from AWS, Azure, and GCP into a central correlation engine.

Micro Focus

Sep 2017 – Jan 2023

Site Reliability Engineer / DevOps Engineer ·

Santa Clara, CA

- Progressed from DevOps Engineer to SRE for OpsBridge Manager (AIOps) and Network Operations Manager; built Kafka and Pulsar telemetry pipelines processing 8TB/day at sub-second latency with end-to-end delivery guarantees into a central data lake.

- Built the Automated Event Correlation (AEC) engine for OpsBridge Manager, grouping related alerts by topology and time proximity to cut alert noise and surface root-cause events faster during high-volume incidents.
- Owned the Terraform module library, reducing environment provisioning time by 70%, and automated product deployments for new customer onboarding, cutting manual setup work by 80%.

Hewlett Packard Enterprise (HPE)

Nov 2015 – Aug 2017

Software Designer ·

Sunnyvale, CA

- Engineered event back-sync modules and product integrations for Operations Bridge Manager; built collectors for key performance indicators across VM technologies (vCenter, KVM).
- Managed technical issues for assigned projects and worked cross-functionally with development teams to analyze system solutions against evolving client requirements, reducing days-to-resolve on customer-encountered defects.

Hewlett Packard (HP)

Sep 2004 – Oct 2015

Software Designer ·

Sunnyvale, CA / Bangalore, IND

- Scaled the Backbone Communication Component from 2,000 to 10,000 concurrent connections in C/C++; built a reverse-channel proxy enabling platform connectivity through a single outbound connection with no inbound firewall rules.
- Engineered embedded Perl memory caching that enabled two years of uninterrupted operation in mission-critical environments; developed a custom log-hashing algorithm that reduced false-positive alerts by 60%.
- Resolved 100% of high-priority on-site escalations at Apple, Samsung, Disney, FedEx, and PG&E through architecture-level root-cause analysis; recipient of HP's Global High Impact Award (2005).

ProcSys

Sep 1999 – Aug 2004

Software Engineer ·

Bangalore, IND

- Built OpsXML, an intent-driven, declarative framework for automating Cisco router configuration. One of the earliest infrastructure-as-code implementations for physical networking.
- Built an eLearning platform for Cisco devices used by thousands of students prepping for the CCNA exam.

RELIABILITY & SCALE IMPACT

- 99.99% uptime sustained as sole SRE owner of a multi-tenant SaaS platform across the Americas, with zero missed SLA commitments over 8 years.
- \$2.1M in annual infrastructure cost cut through Kubernetes consolidation and right-sizing, validated against pre-migration cloud billing baselines.
- 8TB/day telemetry throughput sustained at sub-second delivery latency across Kafka and Pulsar pipelines feeding a real-time correlation engine.
- 100% resolution rate on high-priority on-site escalations across Fortune 500 accounts, each requiring architecture-level root-cause work under live production pressure.

CROSS-FUNCTIONAL COLLABORATION

- Partnered with product, security, and application teams at Wipro/PNC to define SLO targets and provisioning standards adopted across a 70,000-user platform.
- Worked directly with R&D and support leadership at Micro Focus/OpenText, feeding customer and field data back into the product roadmap for 30+ production SaaS deployments.
- Led pre-sales discovery calls and client-facing solution presentations at Kakudi, translating stakeholder requirements into cost and timeline scoping for nonprofit and real-estate clients.

EDUCATION

Master of Science (MS), Software Engineering | BITS Pilani

Bachelor of Engineering (BE), Computer Science & Engineering | University of Mysore

PUBLICATIONS & IP

- "Day One AI: From Your First Job to Your First Product" (Amazon Kindle, ISBN 979-8-9957306-0-6).
- SSRN Author ID 10270516 — papers on AI governance, agent security policy, and fiscal policy for autonomous systems, including "Taxing the AI Agents" (No. 6194458).
- Wipro patent pending: Generation-Bound Cache Invalidation.

ADDITIONAL

- Volunteer AI instructor for high school students; STEM event judge; contributor to Better Programming, dev.to, and Hashnode.
- Cloud Computing certificate, Stanford Continuing Studies.
- ORCID: 0009-0004-9496-6339.